Case 34: Standard fuel-model Rothermel sweep

1 Verification purpose

This controlled known-response case verifies that ELMFIRE reads standard fuel-model parameters and evaluates the no-wind, no-slope Rothermel surface spread model for FBFM2, FBFM9, GR2, TU5, and TL8. It exercises fuel-table parsing, class weighting, reaction intensity, heat sink, and the base rate of spread (ROS) while crown fire, spotting, buildings, acceleration, wind, and slope are excluded. It can expose column-order, unit, weighting, and model-selection errors. This is implementation verification, not comparison with an observed fire. The verification question is whether all five simulation conditions reproduce the independent reaction-intensity and base-ROS calculations within 1% and 2%, with time of arrival (TOA)-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

The no-wind, no-slope rate is

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \tag{1}$$

where I_R is reaction intensity, ξ is the propagating-flux ratio, ρ_b is bulk density, ϵ is effective heating number, and $Q_{ig} = 250 + 1116 M_f$ in native Rothermel units. Dead and live classes are area-weighted before their moisture-damping contributions are combined. The equations and standard fuel definitions follow Patricia L. Andrews, *The Rothermel Surface Fire Spread Model and Associated Developments*, RMRS-GTR-371 (2018). ELMFIRE returns spread rate in metres per minute and reaction intensity in kW m^{-2}.

3 Derivation of expected behavior

The case-specific independent Rothermel reference calculation reads the same frozen fuel-table rows but independently evaluates every algebraic step stated above. It does not import ELMFIRE source or use a simulation output as a reference. Each predicted I_R and R_0 is therefore a deterministic known response. A bounded planar signed-distance front turns the scalar normal speed into a second independent prediction, $t(x) = t_0 + (x - x_0)/(R_0/60)$.

Each reference is calculated independently with

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \qquad Q_{ig} = 250 + 1116 M_f,$$

using the generated common moisture inputs and the named standard-fuel record. The resulting substitutions are

fuel model	R_{ref} (m min^{-1})	$I_{R,\text{ref}}$ (kW m^{-2})
2	0.823796	675.184
9	0.306627	477.714
102	0.228507	186.041
165	0.325722	1595.730
188	0.275244	646.706

(for example, Fuel Model 2 gives 2.702741 ft min$^{-1}(0.3048) = 0.823796$ m min^{-1}). For simulation condition i, direct errors are $e_{R,i} = |R_{\text{max},i} - R_{\text{ref},i}| / \max(|R_{\text{ref},i}|, 10^{-8})$ and the analogous $e_{I,i}$. The

TOA fit is $t = a + bx$, so $R_{\mathrm{TOA}} = 60/b$ and $R^2 = 1 - \sum(t - \hat{t})^2 / \sum(t - \bar{t})^2$. The reported substitutions are $\max_i e_{R,i} \leq 0.02$, $\max_i e_{I,i} \leq 0.01$, $\max_i |R_{\mathrm{TOA},i} - R_{\mathrm{ref},i}|/R_{\mathrm{ref},i} \leq 0.05$, and $\min_i R_i^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

The direct spread raster tests the calculated speed with maximum relative error 0.02; the reaction-intensity raster tests the source term with maximum relative error 0.01. Their medians use colocated valid cells at the planar head, beginning one cell behind the initial east face, so direct verification does not require the front to traverse a downstream corridor. Linear regression of positive arrival time against interior x coordinate tests propagation independently: maximum ROS error is 0.05 and minimum R^2 is 0.995. TOA is required only for simulation conditions whose planned duration supports a multi-cell fit. All five simulation conditions and all four aggregate metrics are required; missing or malformed required evidence gives NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. The domain is 1400 m square with 10 m cells. A finite 50 m-wide ignition strip has a planar east face initially at $x = -400$ m and terminates well inside the north and south boundaries; the interior analysis corridor is therefore one-dimensional without presenting ELMFIRE with an edge-to-edge contained front. Moistures are 5, 7, and 9% for dead classes and 90% for both live classes. Wind and slope are zero. Run duration is selected from the predicted rate to provide approximately 300 m of travel. The interior results should be planar but need not have a monotonic ordering across fuel models because packing, surface-area-to-volume ratio (SAV), load, depth, and extinction moisture interact.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Independent variable
Moisture	dead 5/7/9; live 90/90	percent	Fixed damping state
Wind/slope	0 / 0	mph / degrees	Isolates base fuel response
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

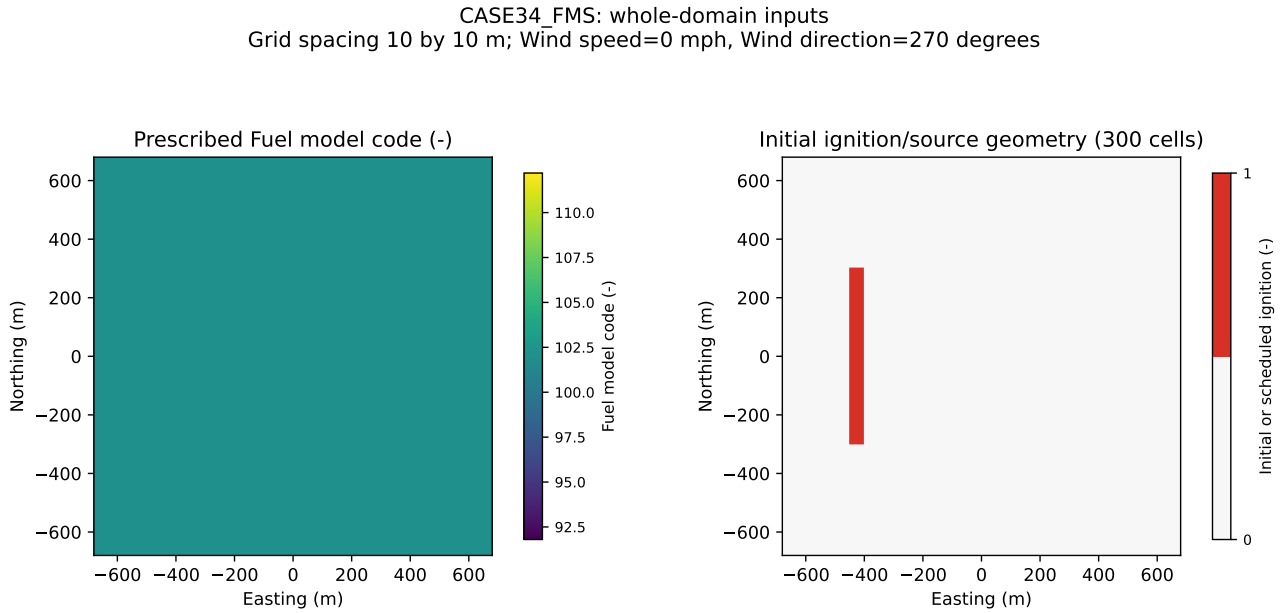


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 5 of 5 required simulation conditions. The scientific decision is **PASS**. No plotted curve is substituted for missing simulation results.

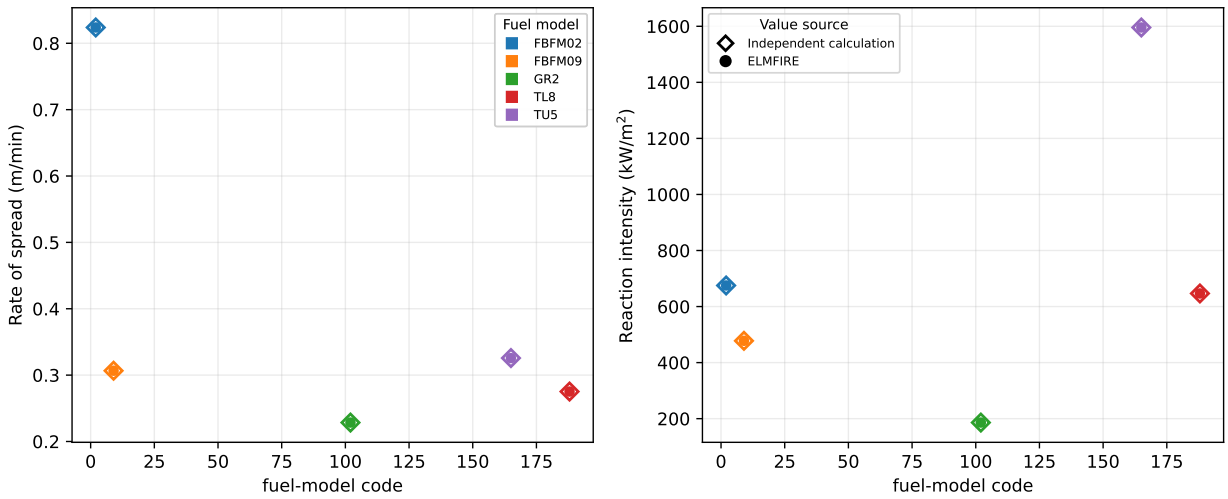


Figure 2: Independently calculated and ELMFIRE rate of spread (ROS, m/min) and reaction intensity (kW/m2) for all five completed standard-fuel simulation conditions. Each fuel model uses one color in both panels; open diamonds denote the independent hand calculation and filled circles denote ELMFIRE.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	2.57611e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0115779 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**; a successful executable exit alone is not a pass.

8 Technical interpretation

An isolated intensity discrepancy implicates fuel parsing, class weighting, or moisture damping. Correct intensity with incorrect direct ROS implicates flux ratio, density, heat sink, adjustment, or output units. Correct direct ROS with a TOA discrepancy implicates level-set transport, raster orientation, or time units.

9 Scope and limitations

The case verifies five representative standard fuels at one moisture state. It does not validate Rothermel against experiments, exhaust all fuel models, or test wind, slope, curing, crown fire, or spatial heterogeneity; those mechanisms have separate cases.